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Student Number

2016/17

EXAMINATION 1

Extension 2 Mathematics

General Instructions

- Reading time – 5 minutes
- Working time - 1 hours
- Write using blue or black pen
Black pen is preferred
- Approved calculators may be used
- The 2 unit / Extension 1 formulae sheet is provided with this paper.
- In Questions 5-6 show relevant mathematical reasoning and/or calculations
- Start a new booklet for each question

Total Marks – 40

Section I - Page 4

4 marks

- Attempt Questions 1 – 4
- Allow about 6 minutes for this section

Section II - Pages 5-8

36 marks

- Attempt Questions 5 – 7
- Allow about 54 minutes for this section

Question	Marks
1 - 4	/4
5	/14
6	/11
7	/11
Total	/40

THIS QUESTION PAPER MUST NOT BE REMOVED FROM THE EXAMINATION ROOM

This assessment task constitutes 15% of the Higher School Certificate Course Assessment

Section I

4 marks

Attempt Questions 1 – 4

Allow about 6 minutes for this section

Use the multiple-choice answer sheet for questions 1 – 4 (Detach from paper)

Q1

What is the value of $\frac{3}{iz}$ if $z = 1 - i$?

(A) $-3 - i$

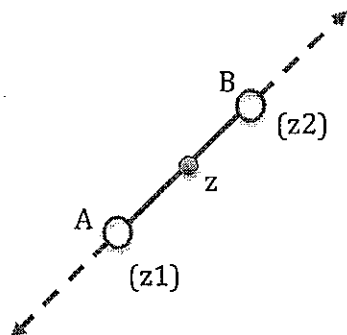
(B) $3 - 3i$

(C) $\frac{-3-3i}{2}$

(D) $\frac{3-3i}{2}$

Q2

The locus of z is the line segment AB, excluding the points A and B.



Which of the following is true?

(A) $\arg(z - z_1) - \arg(z - z_2) = 0$

(B) $\arg(z_2 - z_1) - \arg(z - z_1) = \pi$

(C) $\arg\left(\frac{z-z_1}{z-z_2}\right) = \pi$

(D) $\arg(z_1 - z) = \arg(z_2 - z)$

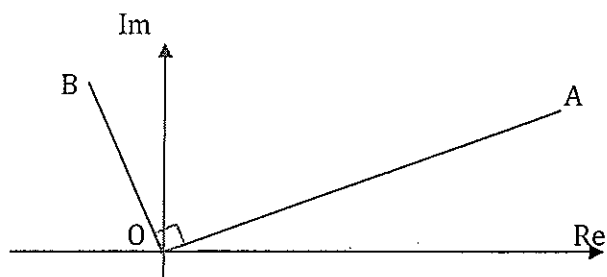
Q3

If $n^2 \equiv an(n+1) + b(n-1) + c$, then the value of the constants a , b and c is:

- (A) $a=1, b=1, c=1$
- (B) $a=1, b=-1, c=1$
- (C) $a=1, b=1, c=-1$
- (D) $a=1, b=-1, c=-1$

Q4

In the Argand diagram below, the points A and B represent the complex numbers z and ω respectively, where $\angle AOB=90^\circ$.



If $|\overrightarrow{OA}| = 2|\overrightarrow{OB}|$ which of the following is correct?

- (A) $\omega = 2iz$
- (B) $\omega = -2iz$
- (C) $\omega = -\frac{iz}{2}$
- (D) $\omega = -\frac{z}{2i}$

Section II

36 marks

Attempt Questions 5-7

Allow about 54 minutes for this section

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.

In Questions 4 – 5, your responses should include relevant mathematical reasoning and/or calculations.

Question 5 (14 marks) Use a SEPARATE writing booklet

(a)	The polynomial equation $x^3 + 3x^2 - 2x - 1 = 0$ has roots α, β, γ . Find the polynomial equation whose roots are $2\alpha, 2\beta, 2\gamma$?	2
(b)	A polynomial equation $P(x)$ has remainder 6 when divided by $x + 2$, and remainder 9 when divided by $x - 1$. What is the remainder when $P(x)$ is divided by $(x + 2)(x - 1)$?	3
(c)	The polynomial equation $P(x) = ax^4 + bx^3 + cx^2 + e$ has remainder -2 when divided by $x + 1$ and a double root at $x = 1$. Show that $4a + 2c = -3$	3
(d)	Let $P(z) = z^4 + 2kz^3 + 2k^2z^2 + 2kz + 1$, where k is real. Let $\beta = x + iy$, where x and y are real. Suppose that β and $i\beta$ are zeros of $P(z)$, where $\bar{\beta} \neq i\beta$.	
	i) Explain why $\bar{\beta}$ and $-i\bar{\beta}$ are zeros of $P(z)$	1
	ii) Show that $P(z) = z^2(z + k)^2 + (kz + 1)^2$	1
	iii) Hence show that if $P(z)$ has a real zero then $P(z) = z^2(z - 1)^2 + (1 - z)^2$ OR $P(z) = (z^2 + 1)(z + 1)^2$.	2
	iv) Show that all zeros of $P(z)$ have modulus 1.	2

End of Question 5

Question 6 (11 marks) Use a SEPARATE writing booklet

(a)	i) Find the square roots of $-8 + 6i$.	2
	ii) Solve $z^2 - (3 - 5i)z - 2 - 9i = 0$.	2
	iii) Sketch the locus of z , if $4 z - 2 + i = z^2 - (3 - 5i)z - 2 - 9i $	3
(b)	(i) Solve the equation $z^5 + 32 = 0$ and list down all the roots in mod-arg form.	2
	(ii) Hence express $z^4 - 2z^3 + 4z^2 - 8z + 16$ as the product of two quadratic factors with real coefficients.	2
End of Question 6		

Question 7 (11 marks) Use a SEPARATE writing booklet

(a)

Points A, B, C and D on an Argand diagram representing the complex numbers $-2i$, 4 , $2 + 4i$ and $2i$ respectively. ABCD form a convex quadrilateral as shown below. Squares are drawn outside the quadrilateral on each of the four sides. P, Q, R and S are the centres of the squares on AB, BC, CD and AD respectively.

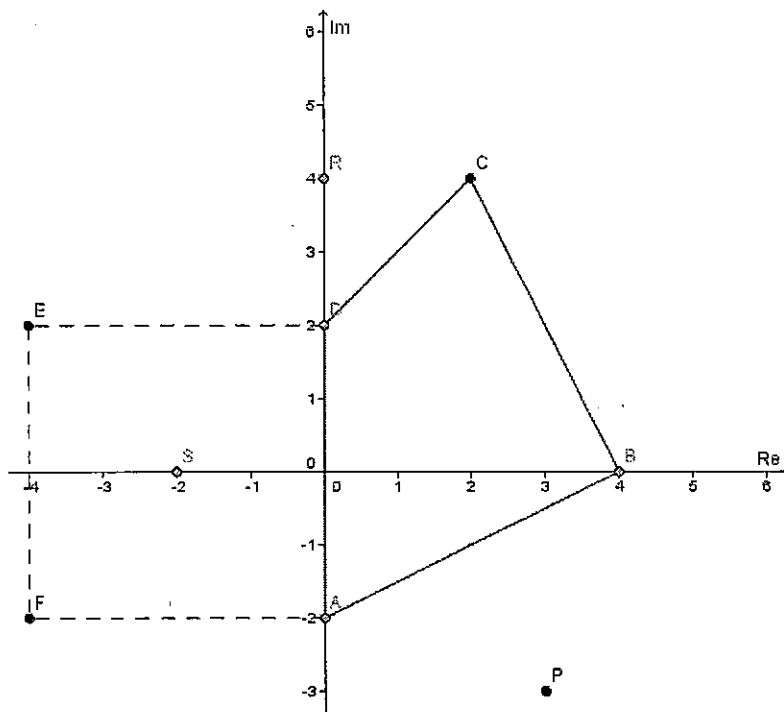
- i) Using vector representation of complex numbers, show that the complex number represented by Q is $5 + 3i$.

3

(It is given that P, R and S represent $3 - 3i$, $4i$ and -2 respectively. You don't need to prove this)

- ii) Hence prove that PR is equal and perpendicular to QS, using vector representation of complex numbers.

3



(b) If $z = \cos\theta + i\sin\theta$,

- (i) Show that $z^n + \frac{1}{z^n} = 2\cos(n\theta)$ and $z^n - \frac{1}{z^n} = 2i\sin(n\theta)$.

2

- (ii) Hence show that $\cos^4\theta + \sin^4\theta = \frac{1}{4}(\cos 4\theta + 3)$

3

End of Question 7

End of Examination ☺



Student Number

1. A ☐ B ☐ C ☐ D ☐
2. A ☐ B ☐ C ☐ D ☐
3. A ☐ B ☐ C ☐ D ☐
4. A ☐ B ☐ C ☐ D ☐

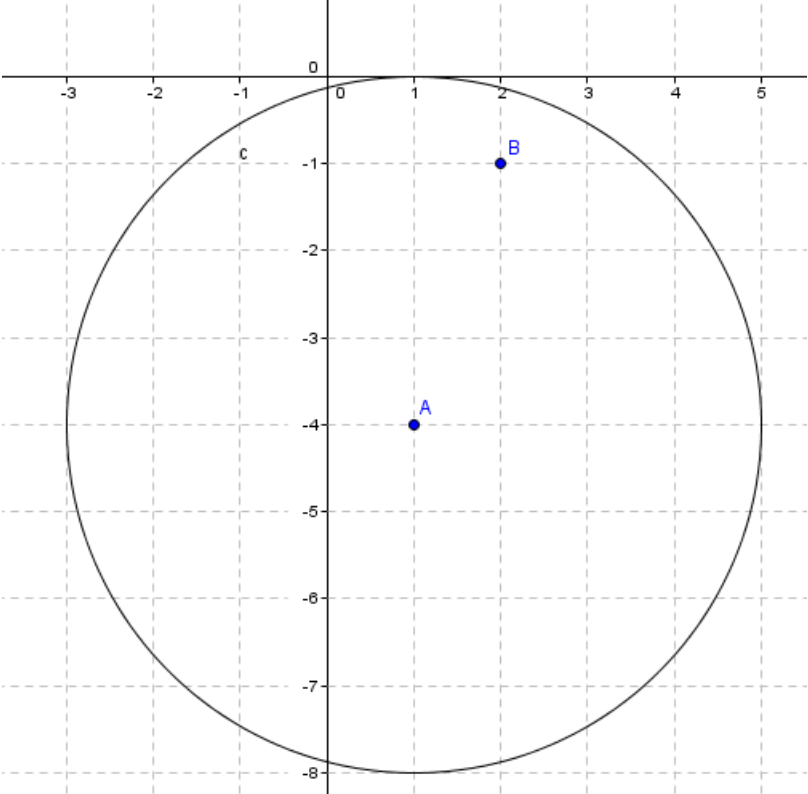
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Extension 2 Task 1 2017 ANSWERS

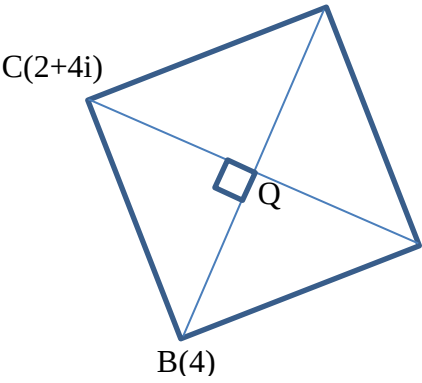
Q	Answer	Marks
1	D	1
2	C	1
3	D	1
4	D	1
5	a) Let $P(x) = x^3 + 3x^2 - 2x - 1 = 0$ have roots α, β, γ. α is a root of $P(x) \rightarrow P(\alpha) = 0$ When $2\alpha = x$, $\alpha = \frac{x}{2} \rightarrow P\left(\frac{x}{2}\right) = 0$ is the polynomial equation whose roots are $2\alpha, 2\beta, 2\gamma$ $P\left(\frac{x}{2}\right) = \left(\frac{x}{2}\right)^3 + 3\left(\frac{x}{2}\right)^2 - 2\left(\frac{x}{2}\right) - 1 = 0$ i.e. $x^3 + 6x^2 - 8x - 8 = 0$ b) When $P(x)$ is divided by $(x+2)$ the remainder is $P(-2)$ by remainder theorem. $\therefore P(-2) = 6$ Similarly $P(1) = 9$ Since $(x+2)(x-1)$ is a quadratic polynomial, the remainder when $P(x)$ is divided by $(x+2)(x-1)$ is a linear function $(ax+b)$ and let the quotient function be $Q(x)$. $\therefore P(x) \equiv (x+2)(x-1).Q(x) + (ax+b)$ This division transformation is an identity i.e. the statement is true for all values of x. $\therefore$ when $x = -2 \quad P(-2) = -2a + b = 6 \quad (1)$ $x = 1 \quad P(1) = a + b = 9 \quad (2)$ Solving equations (1) and (2) simultaneously gives $a = 1, b = 8$ $\therefore$ the required remainder is $(x+8)$	1 1 1 1 1

Q	Answer	Marks
5	c) By remainder theorem when $P(x)$ is divided by $(x + 1)$ the remainder is $P(-1)$. $P(-1) = -2 \rightarrow a(-1)^4 + b(-1)^3 + c(-1)^2 + e = -2$ $\rightarrow a - b + c + e = -2 \quad (1)$ $P(x)$ has a double root at $x = 1 \rightarrow$ $P(1) = 0 \rightarrow a + b + c + e = 0 \quad (2)$ And as $P'(x) = 4ax^3 + 3bx^2 + 2cx = 0$ $P'(1) = 0 \rightarrow 4a + 3b + 2c = 0 \quad (3)$ $(2) - (1) \rightarrow 2b = 2 \rightarrow b = 1$ Sub $b = 1$ into (3) $4a + 3 + 2c = 0$ $4a + 2c = -3$ d) i) Complex conjugate of $\beta = \bar{\beta}$ Complex conjugate of $i\bar{b} = \overline{i\bar{b}} = i\bar{b} = -i\bar{b}$ $P(z) = z^4 + 2kz^3 + 2k^2z^2 + 2kz + 1$, where k is real, and the coefficients of $P(z)$ are real $\rightarrow \beta$ is a root, implies $\bar{\beta}$ is a root of $P(z)$ by conjugate root theorem Similarly $i\beta$ is a root, implies $-i\bar{\beta}$ is a root of $P(z)$ ii) $P(z) = z^4 + 2kz^3 + 2k^2z^2 + 2kz + 1$ $= (z^4 + 2kz^3 + k^2z^2) + (k^2z^2 + 2kz + 1)$ $= z^2(z + k)^2 + (kz + 1)^2$ iii) $P(z) = z^2(z + k)^2 + (kz + 1)^2$ As $P(z)$ is the sum of two complete squares, for $P(z)$ to be zero, both $z + k$ and $kz + 1$ must be zero. Solving $z + k = 0$ and $kz + 1 = 0$ gives $k = \pm 1$. When $k = 1, P(z) = z^2(z + 1)^2 + (z + 1)^2$ $= (z + 1)^2(z^2 + 1) \text{ ---- (1) (by factorising)}$ When $k = -1, P(z) = z^2(z - 1)^2 + (1 - z)^2 \text{ ---- (2)}$ iv) From (iii) above, when $P(z) = 0$, gives $z = -1$ and $\pm i$ from (1) and $z = 1$ from (2) ie. The roots of $P(z) = 0$ are $1, -1, i$ and $-i$ The modulus of each of these roots are 1.	1 1 mark for using $P(1) = 0$ and $P'(1) = 0$ 1 mark for using the equations to get the required answer 1 mark for stating the conjugate theorem and that all coefficients are real 1 mark for correct working 1 mark for getting the two values of k 1 mark for getting the two required expressions for $P(z)$. 1 mark for each for using the two equations to get the values of z 1 mark for stating the modulus is 1

Q	Answer	Marks
6	a) i) Students may do by a number of methods (by calculating using shorthand techniques) Go straight to $-8 + 6i = (1 + 3i)^2$ $\therefore \sqrt{-8 + 6i} = \pm(1 + 3i)$ OR long way $\sqrt{-8 + 6i} = x + iy$ $\therefore -8 + 6i = x^2 - y^2 + 2ixy$ Equating coefficients of real and imaginary parts $-8 = x^2 + y^2 \quad (1)$ $6i = 2ixy$ $3 = xy$ $y = \frac{3}{x} \quad (2)$ Sub. (2) into (1) $-8 = x^2 - \left(\frac{3}{x}\right)^2$ $-8x^2 = x^4 - 9$ $x^4 + 8x^2 - 9$ $(x^2 + 9)(x^2 - 1) = 0$ $\therefore x = \pm 1$ sub. into (2) $y = \pm 3$ $\therefore \sqrt{-8 + 6i} = \pm(1 + 3i)$	2 marks for getting the two square roots correctly by any relevant method

Q	Answer	Marks
6	a/ ii) $z^2 - (3 - 5i)z - 2 - 9i = 0$ By quadratic formula $z = \frac{(3 - 5i) \pm \sqrt{(3 - 5i)^2 + 4(2 + 9i)}}{2}$ $z = \frac{(3 - 5i) \pm \sqrt{-8 + 6i}}{2}$ $z = \frac{(3 - 5i) \pm (1 + 3i)}{2} \quad \text{from part i)}$ $z = (2 - i), (1 - 4i)$ iii) $4 z - 2 + i = z^2 - (3 - 5i)z - 2 - 9i$ $\rightarrow 4 z - 2 + i = [z - (2 - i)][z - (1 - 4i)] \quad \text{from part ii}$ $\rightarrow 4 z - 2 + i = z - 2 + i z - 1 + 4i$ $\rightarrow z - 2 + i (z - 1 + 4i - 4) = 0$ When $z - 2 + i = 0 \rightarrow$ the locus of z is just the point $(2 - i)$ When $z - 1 + 4i = 4 = 0 \rightarrow$ the locus of z is the circle with centre $(1 - 4i)$ and radius 4. Sketch is shown below: 	2 1 1 1

Q	Answer	Marks
6	b) i) $z^5 = -32$ $z^5 = 2^5(-1 + 0i)$ $z^5 = 2^5 \operatorname{cis}(2k - 1)\pi \text{ where } k \text{ is an integer}$ $z = 2 \operatorname{cis}(2k - 1)\frac{\pi}{5} \text{ (By De Moivre's Theorem)}$ The roots of $z^5 + 32 = 0$ are: when $k = 0$, $z = 2 \operatorname{cis}\left(-\frac{\pi}{5}\right) = z_1$ when $k = 1$, $z = 2 \operatorname{cis}\left(\frac{\pi}{5}\right) = \bar{z}_1$ when $k = -1$, $z = 2 \operatorname{cis}\left(-\frac{3\pi}{5}\right) = z_2$ when $k = 2$, $z = 2 \operatorname{cis}\left(\frac{3\pi}{5}\right) = \bar{z}_2$ when $k = -2$, $z = 2 \operatorname{cis}(\pi) = -2$ ii) $\frac{z^5 + 32}{z + 2} = \frac{z^5 + 32}{z + 2} = [z^2 - (z_1 + \bar{z}_1)z + z_1 \bar{z}_1][z^2 - (z_2 + \bar{z}_2)z + z_2 \bar{z}_2]$ $\frac{z^5 + 32}{z + 2} = \left[z^2 - 4 \cos \frac{\pi}{5} z + 4\right] \left[z^2 - 4 \cos \frac{3\pi}{5} z + 4\right]$	2 for all correct roots 1 for factorising and grouping the conjugates 1 for the final answer

Q	Answer	Marks
7	a)  i) In right angled ΔBQC, $\frac{CQ}{CB} = \cos \frac{\pi}{4}$ $ie. CQ = \frac{1}{\sqrt{2}} CB$ $\overrightarrow{CQ}$ is obtained by rotating $\frac{1}{\sqrt{2}} \overrightarrow{CB}$ in the anti-clockwise sense by $\frac{\pi}{4}$ radians about C. $\therefore \overrightarrow{CQ} = \frac{1}{\sqrt{2}} \overrightarrow{CB} \text{ cis } \frac{\pi}{4}$ $\overrightarrow{CB} = \overrightarrow{CO} + \overrightarrow{OB}$ $= (-2 - 4i) + 4$ $= 2 - 4i$ $\therefore \overrightarrow{CQ} = \frac{1}{\sqrt{2}} (2 - 4i) \text{ cis } \frac{\pi}{4}$ $= (1 - 2i)(1 + i)$ $= 3 - i$ $\overrightarrow{OQ} = \overrightarrow{OC} + \overrightarrow{CQ}$ $= (2 + 4i) + (3 - i)$ $= (5 + 3i)$ $\therefore Q$ represents $(5 + 3i)$. ii) $\overrightarrow{PR} = \overrightarrow{PO} + \overrightarrow{OR}$ $= (-3 + 3i) + 4i$ $= -3 + 7i$ $\overrightarrow{QS} = -2 - (5 + 3i)$ $= (-7 - 3i)$ $i\overrightarrow{PR} = i(-3 + 7i)$ $= (-7 - 3i)$ $= \overrightarrow{QS}$	1 1 1 1

	Since $\overrightarrow{QS} = i\overrightarrow{PR}$, $\overrightarrow{QS} = i\overrightarrow{PR}$ and $\arg(\overrightarrow{QS}) = \arg(i\overrightarrow{PR})$. $\overrightarrow{QS} = i\overrightarrow{PR}$ $ \overrightarrow{QS} = i \overrightarrow{PR} $ $= \overrightarrow{PR} $ ie. the lengths of QS and PR are equal. $\arg(\overrightarrow{QS}) = \arg(i\overrightarrow{PR})$ $\arg(\overrightarrow{QS}) = \arg(i) + \arg(\overrightarrow{PR})$ $= \frac{\pi}{2} + \arg \overrightarrow{PR}$ ie. QS and PR are perpendicular Therefore QS and PR are both equal and perpendicular.	1 1
7	b (i) $z = \cos\theta + i\sin\theta$ $z^n = \cos(n\theta) + i\sin(n\theta) \quad (1) \quad \text{By De Moivre's Theorem}$ $\frac{1}{z^n} = z^{-n}$ $= \cos(-n\theta) + i\sin(-n\theta) \quad \text{By De Moivre's Theorem}$ $= \cos(n\theta) - i\sin(n\theta) \quad (2)$ (1) + (2) gives $z^n + \frac{1}{z^n} = 2\cos(n\theta)$ (1) - (2) gives $z^n - \frac{1}{z^n} = 2i\sin(n\theta)$	1 Mark for using De Moivre's theorem Correctly 1 mark for getting the expressions for $z^n + \frac{1}{z^n}$ and $z^n - \frac{1}{z^n}$

	ii) $2 \cos \theta = z + \frac{1}{z}$ $\therefore 16 \cos^4 \theta = \left(z + \frac{1}{z} \right)^4 \quad (1)$ $2i \sin \theta = z - \frac{1}{z}$ $\therefore 16 \sin^4 \theta = \left(z - \frac{1}{z} \right)^4 \quad (2)$ $(1)+(2) \quad 16(\cos^4 \theta + \sin^4 \theta) = \left(z + \frac{1}{z} \right)^4 + \left(z - \frac{1}{z} \right)^4$ $= 2 \left(z^4 + \frac{1}{z^4} \right) + 12$ $= 2 \cos 4\theta + 12$ $\therefore \cos^4 \theta + \sin^4 \theta = \frac{1}{8} [\cos 4\theta + 3]$	1 mark for getting expressions for $\cos^4 \theta$ and $\sin^4 \theta$ 1 mark for simplifying 1 mark for using De Moivre's theorem again to get the answer
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